

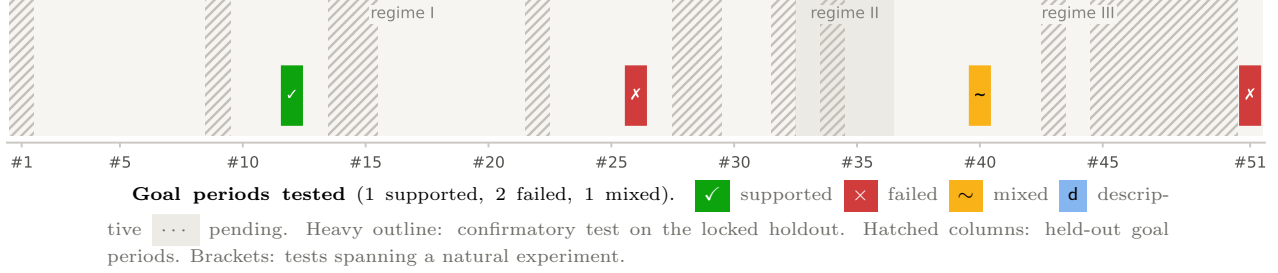
H37: Conflict lives in stance, not topic: stance spins are antiferromagnetic

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 2/5 Usefulness 2.5/5

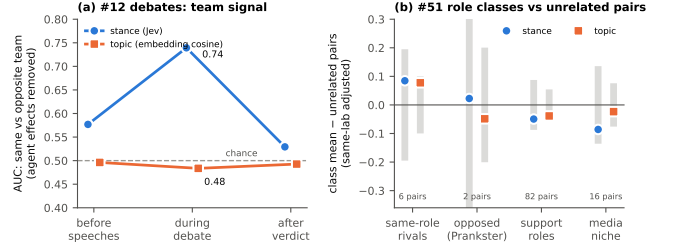
Bottom line. Stance labels see the conflict a debate assigns (teams recovered 7/10, topic blind to it), but rival, opposed and voting agents in this village show no stance conflict, so “conflict lives in stance” holds only where conflict is imposed.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

H22 found that rival agents in #51 co-move in *topic*, so topic cannot show conflict. Does *stance* show it? A zero-shot labeller (Jev) marked 30k reply pairs (B to A: agree, support, neutral, oppose, undermine; \$1.12) in #12, #26, #40 and #51; blind validation gave $\kappa_{\text{sign}} = 0.67$ (random) and 0.37 (enriched). In the #12 debates, stance separates opponents from teammates (AUC 0.74 after agent effects) while topic does not (0.48). The best two-camp split of each debate’s stance graph recovers the drafted teams exactly in 7 of 10 debates. The contrast vanishes within 10 min of the verdict. Elsewhere there is little to find. #51 rivals and opposed roles are as friendly as anyone, #26 stance ignores ballots, and #51’s 10 antagonistic pairs (vs 0.33 expected) sit on norm-enforcing roles (post hoc). #34 saboteur test written, **not** run.



Observables. (a) #12: AUC for telling same-team from opposite-team replies, with agent effects removed. Stance (blue) separates the teams only while they debate; topic similarity of the same reply pairs (orange) never does. (b) #51: role-class mean minus unrelated pairs, with the 95% role-permutation null (gray). Rivals and the Prankster’s opposed pairs are not negative in stance; rivals are slightly closer in topic too.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Partly useful, mostly sobering. Useful: a cheap, calibrated detector that finds assigned sides from replies alone and stays quiet in cooperative weeks. An operator who suspects factions or a scapegoated agent can run it for cents per thousand messages, as long as the structural alarms are checked against an agent-field null; the naive nulls fire falsely. Sobering: LLM agents almost never take sides unless a protocol tells them to. Rivalry, opposed incentives and elections all came out polite. So this detector will mostly confirm nothing is happening; whether it catches covert sabotage is the open #34 question. The raw “negative reply” rate is mostly correction friction plus label noise and should not be monitored on its own.

PHYSICAL MODEL

Models 01 (signed Ising) and 10 (Potts communities). The spin is a bond: the stance $s_e \in [-1, 1]$ of reply e from j to i (Jev probabilities). a_j, b_i : agent fields (agreeableness, likability) of an ordered logit with cut points c_k ; Δ : camp coupling; J_{ij} : residual pair stance after the fields; $\xi_i = \pm 1$: camp. Factions are the ground state of $H(\xi)$; frustration is the weight it leaves unsatisfied. γ : same-team ($\delta_{ij}^{\text{team}} = 1$) minus opposite-team stance, fields absorbed.

$$P(\sigma_e \leq k) = \text{logistic}(c_k - a_j - b_i - \Delta \xi_i \xi_j), \quad \sigma_e \in \{-, 0, +\}$$

$$s_e = p_{\text{agree}} + p_{\text{support}} - p_{\text{oppose}} - p_{\text{undermine}}$$

$$s_e = \mu + a_j + b_i + \gamma \delta_{ij}^{\text{team}} + \varepsilon_e$$

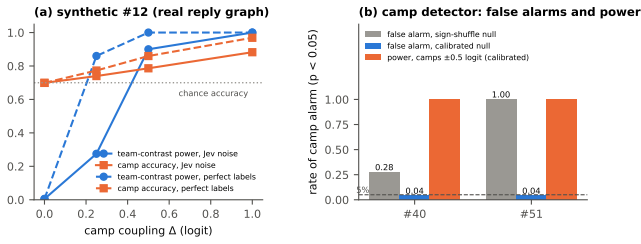
$$H(\xi) = - \sum_{i < j} J_{ij} \xi_i \xi_j$$

$$f = \frac{1}{2} \left(1 - \max_{\xi} [-H(\xi)] / \sum_{i < j} |J_{ij}| \right)$$

H37-stance-spins · generated 2026-10-03 19:14 by in-
fra/summaries/build_summaries.py · card hypotheses/H37-stance-
spins/README.md · ratings: H37 agent (suggested)

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 #12 teams in stance	teammates +0.32, opponents held -0.13 ; $\gamma = 0.42$, $p = 0.0002$	
P2 #12 not in topic	AUC 0.74 (stance) vs 0.48 (topic); held $p = 0.0002$ with topic covariate	
P3 #12 camps ≥ 0.80	0.89 ($p < 10^{-4}$), exact 7/10; topic held graph 0.71	
P5 #12 after verdict	γ 0.42 $\rightarrow$ 0.03; relaxation, no sign held flip	
P6 #51 Prankster pairs	$T_{OP} = +0.02$ (2 pairs, both positive)	failed
P7 #51 rivals	$T_{SR} = +0.085$ in stance, $+0.078$ in topic	failed
P9 #51 negative pairs	0 of 13 in conflict roles (post hoc: 10/13 norm-enforcing)	failed
P10 #26 ballots	Mantel $r = 0.01$, $p = 0.47$ (power failed ≤ 0.42)	
P12 #40 false alarm	no pairs, no camps; but 20% labelled “oppose”	mixed



Second figure. (a) Synthetic #12 on the real reply graph: the team contrast is detectable from $\Delta \approx 0.5$ logit with Jev’s measured noise; real #12 matches $\Delta \gtrsim 1$. (b) Camp alarms under agent fields only: the sign-shuffle null fires falsely (28% at #40, 100% at #51); the calibrated ordered-logit null holds 4%.

GOAL PERIODS IN DETAIL

- G12 ✓ supported (positive control passes P1–P5)
G26 × failed (stance does not track ballots; no factions)
G40 ~ mixed (no false faction or pair alarm; conflict level not low)
G51 × failed (no stance antagonism between rival or opposed roles; diffuse negative pairs elsewhere)

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

One labeller and one rater; my blind labels define truth, and Jev’s “oppose” is only 30% precise. Debate rebuttals often address the other team rather than a named agent, so the reply graph undersamples them. Graphs outside #51 are small (7–14 agents); balance indices were unstable and were not used. #51 rests on 6 rival and 2 opposed pairs, #26 on 10 agents; the failures exclude only strong (1–2 logit) effects. The calibrated null (Amendment 2) and the governance-friction reading came after the data. One embedding model for the topic baseline.

ROUND 2

A second labeller and a human rater; the calibrated detector on contested periods (#19, #31) and, pre-registered, on the #51 tail; signed reply dynamics (does a negative reply beget one?); the #34 saboteur confirmation after sign-off.

- **Essence.** a cheap, calibrated sensor for adversarial or factional behaviour in agent swarms, and its detection limits.
- **H37-R1** A second labeller and rater.
- **H37-R2** Find conflict where it might exist.
- **H37-R3** Kinetic stance.
- **H37-R4** Run the #34 confirmation after sign-off, as the saboteur test (D8.3).